Light

GCSE Computer Science — Programming Testing and Robustness

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a syntax error?

- ○ **A.** An error that occurs when the program runs out of memory
- ○ **B.** An error that causes the program to produce incorrect results
- ○ **C.** An error where the code breaks the rules of the programming language
- ○ **D.** An error that only appears when specific data is entered

Q2 (1 mark)

What is a logic error?

- ○ **A.** An error where the code runs but produces incorrect output
- ○ **B.** An error where the program crashes unexpectedly
- ○ **C.** An error caused by a missing semicolon or bracket
- ○ **D.** An error detected by the compiler before running

Q3 (1 mark)

Which of the following is an example of a runtime error?

- ○ **A.** Forgetting to close a bracket
- ○ **B.** Using the wrong variable name that doesn't exist
- ○ **C.** Dividing a number by zero
- ○ **D.** Writing 'pirnt' instead of 'print'

Q4 (1 mark)

What is normal test data?

- ○ **A.** Data that is outside the expected range
- ○ **B.** Data that should be rejected by the program
- ○ **C.** Data that tests the limits of acceptable values
- ○ **D.** Data that is typical and should be accepted by the program

Q5 (1 mark)

A program accepts values between 1 and 100. Which value would be considered boundary test data?

- ○ **A.** 50
- ○ **B.** 100
- ○ **C.** 150
- ○ **D.** -5

Q6 (1 mark)

What is the purpose of erroneous test data?

- ○ **A.** To check that the program handles invalid input appropriately
- ○ **B.** To verify the program produces correct results
- ○ **C.** To test values at the edge of the valid range
- ○ **D.** To confirm the program runs without errors

Q7 (1 mark)

What type of validation check ensures a value falls within acceptable limits?

- ○ **A.** Presence check
- ○ **B.** Length check
- ○ **C.** Range check

- ○ **D.** Pattern check

Q8 (1 mark)

A form requires users to enter their email address. Which validation check ensures the field is not left empty?

- ○ **A.** Range check
- ○ **B.** Pattern check
- ○ **C.** Length check
- ○ **D.** Presence check

Q9 (1 mark)

Why are comments added to program code?

- ○ **A.** To make the program run faster
- ○ **B.** To explain what the code does for future reference
- ○ **C.** To fix syntax errors in the code
- ○ **D.** To reduce the file size of the program

Q10 (1 mark)

Which variable name follows good programming practice?

- ○ **A.** x

- B. a1
- C. totalScore
- D. ts

Q11 (1 mark)

What is the purpose of authentication in a program?

- A. To check that data is in the correct format
- B. To encrypt data before storage
- C. To validate that input is within range
- D. To verify the identity of a user

Q12 (1 mark)

What is a trace table used for?

- A. To track the values of variables as a program executes
- B. To store user passwords securely
- C. To list all the syntax errors in a program
- D. To record the time taken for each function to run

Q13 (2 marks)

Consider the following Python code:

```
if score > 10
    print("High score")
```

What type of error does this code contain?

- ○ **A.** Logic error
- ○ **B.** Runtime error
- ○ **C.** Syntax error
- ○ **D.** No error

Q14 (2 marks)

A program should calculate the average of three test scores. Consider the following code:

```
score1 = 80
score2 = 70
score3 = 90
average = score1 + score2 + score3 / 3
print(average)
```

This code outputs 180.0 instead of 80.0. What type of error is this?

- ○ **A.** Syntax error
- ○ **B.** Runtime error
- ○ **C.** Type error
- ○ **D.** Logic error

Q15 (2 marks)

Which code correctly implements a range check for age between 18 and 65?

- A. `if age >= 18 and age <= 65:`
- B. `if age > 18 and age < 65:`
- C. `if age > 18 or age < 65:`
- D. `if age == 18 or age == 65:`

Q16 (2 marks)

A password must be at least 8 characters long. Which code correctly validates this?

- A. `if len(password) > 8:`
- B. `if len(password) >= 8:`
- C. `if password >= 8:`
- D. `if len(password) == 8:`

Q17 (2 marks)

A program validates that a quantity is between 1 and 50 (inclusive). Which set of values would be the most appropriate boundary test data?

- A. 25, 26
- B. 0, 1, 50, 51
- C. 10, 20, 30, 40

- **D.** -1, 100

Q18 (2 marks)

What is defensive programming?

- **A.** Writing code that attacks malicious users
- **B.** Writing code using only secure programming languages
- **C.** Writing code that hides error messages from users
- **D.** Writing code that anticipates and handles potential errors

Q19 (2 marks)

Consider the following code:

```
numbers = [10, 20, 30]
print(numbers[5])
```

What type of error will occur when this code runs?

- **A.** Syntax error
- **B.** Logic error
- **C.** Runtime error
- **D.** No error will occur

Q20 (2 marks)

Why is consistent indentation important in programming?

- ○ **A.** It shows the structure of the code and improves readability
- ○ **B.** It makes the program run faster
- ○ **C.** It reduces the memory used by the program
- ○ **D.** It prevents all types of errors

Q21 (2 marks)

A UK postcode should match the pattern of letters and numbers (e.g., "SW1A 1AA"). What type of validation check would verify this format?

- ○ **A.** Range check
- ○ **B.** Presence check
- ○ **C.** Length check
- ○ **D.** Pattern check

Q22 (2 marks)

Consider the following authentication code:

```
stored_password = "secret123"
entered = input("Enter password: ")
if entered == stored_password:
    print("Access granted")
else:
    print("Access denied")
```

What security weakness does this code have?

- ○ **A.** The password is not long enough
- ○ **B.** The password is stored in plain text
- ○ **C.** The code doesn't use a loop
- ○ **D.** The comparison operator is wrong

Q23 (2 marks)

What is the purpose of a code review?

- ○ **A.** To automatically fix all bugs in the code
- ○ **B.** To measure how fast the code executes
- ○ **C.** To convert code from one language to another
- ○ **D.** To have other programmers examine code for errors and improvements

Q24 (3 marks)

A program accepts a percentage mark between 0 and 100 inclusive. Which set of test data provides the most comprehensive testing?

- ○ **A.** -1 (erroneous), 0 (boundary), 50 (normal), 100 (boundary), 101 (erroneous)
- ○ **B.** 50 (normal)
- ○ **C.** 0, 25, 50, 75, 100 (all normal)
- ○ **D.** abc, -100, 200 (all erroneous)

Q25 (3 marks)

The following code should output the sum of numbers from 1 to 5 (which is 15), but it outputs 10:

```
total = 0
for i in range(1, 5):
    total = total + i
print(total)
```

What is the error and how should it be fixed?

- ○ **A.** Syntax error: change 'range(1, 5)' to 'range[1, 5]'
- ○ **B.** Runtime error: change 'total = 0' to 'total = 1'
- ○ **C.** Logic error: change 'total + i' to 'total * i'
- ○ **D.** Logic error: change 'range(1, 5)' to 'range(1, 6)'

Q26 (3 marks)

Consider the following code that should repeatedly ask for a valid age until one is entered:

```
age = int(input("Enter age: "))
while age < 0 or age > 120:
    print("Invalid age")
    age = int(input("Enter age: "))
print("Age accepted:", age)
```

What happens if the user enters "abc" on the first attempt?

- ○ **A.** The program outputs "Invalid age" and asks again
- ○ **B.** The program crashes with a runtime error
- ○ **C.** The program accepts "abc" as a valid age

- ○ **D.** The program outputs "Age accepted: abc"

Q27 (3 marks)

Consider the following code:

```
x = 5
y = 3
while x > 0:
    x = x - 1
    y = y + 2
```

What are the final values of x and y after the loop completes?

- ○ **A.** x = 0, y = 13
- ○ **B.** x = 1, y = 11
- ○ **C.** x = 0, y = 3
- ○ **D.** x = -1, y = 15

Q28 (3 marks)

A password must be at least 8 characters long AND contain at least one digit. Which code correctly validates both requirements?

- ○ **A.** `if len(password) >= 8 or any(c.isdigit() for c in password):`
- ○ **B.** `if len(password) >= 8 and any(c.isdigit() for c in password):`
- ○ **C.** `if len(password) > 8 and password.isdigit():`
- ○ **D.** `if len(password) >= 8 and password.isalpha():`

Q29 (3 marks)

A search algorithm is tested with a list of 1000 items. The target item is found after 500 comparisons on average. When the list grows to 2000 items, the average comparisons increase to 1000. What does this indicate about the algorithm?

- ○ **A.** The algorithm has a logic error
- ○ **B.** The algorithm is a binary search with O(log n) efficiency
- ○ **C.** The algorithm is a linear search with O(n) efficiency
- ○ **D.** The algorithm has a runtime error

Q30 (3 marks)

Consider the following robust input validation code:

```
valid = False
while not valid:
    try:
        age = int(input("Enter age: "))
        if 0 <= age <= 120:
            valid = True
        else:
            print("Age must be 0-120")
    except ValueError:
        print("Please enter a number")
```

Why is this code considered robust?

- ○ **A.** It uses a while loop instead of a for loop
- ○ **B.** It uses the variable name 'valid' which is meaningful
- ○ **C.** It prints messages to the screen

○ **D.** It handles both type errors and range errors gracefully

Total: **55** marks • Time: **30 min**

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